Analytics Programming with R
Chapter 5

Data Frames

- Data Frames are created using the "data.frame()" function
- Use "nrow()" to check the number of rows in a data frame
- Use "ncol()" to check the number of columns in a data frame
- Use "dim()" to check the number of rows and columns in a data frame
- Use "names()" to look at the column names of a data frame
- Use "rownames()" to check the names of each row.
- Use rownames() <- c("One","Two","Three", ect) to assign names to the individual rows in a data frame

Data Frames

- To look at the first few rows of a data frame use the "head()" function.
- To look at the last few rows of a data frame use the function "tail()"
- You can use the "class()" function to check the class of a data frame.
- The $ or [] is used to access individual columns of a data frame.
  - Ex. theDataFrame$Currency
  - Ex. theDataFrame[3, 2] # the row is the first argument and the column is the second argument
- To access multiple columns by name make the column argument a character vector of names
  - theDataFrame[, c("Currency", "ExchRate")]

Data Frames

- When using the [] there is a third argument that can be used to ensure you return a single column, "drop = FALSE"
  - theDataFrame[, "ExchRate", drop = FALSE]

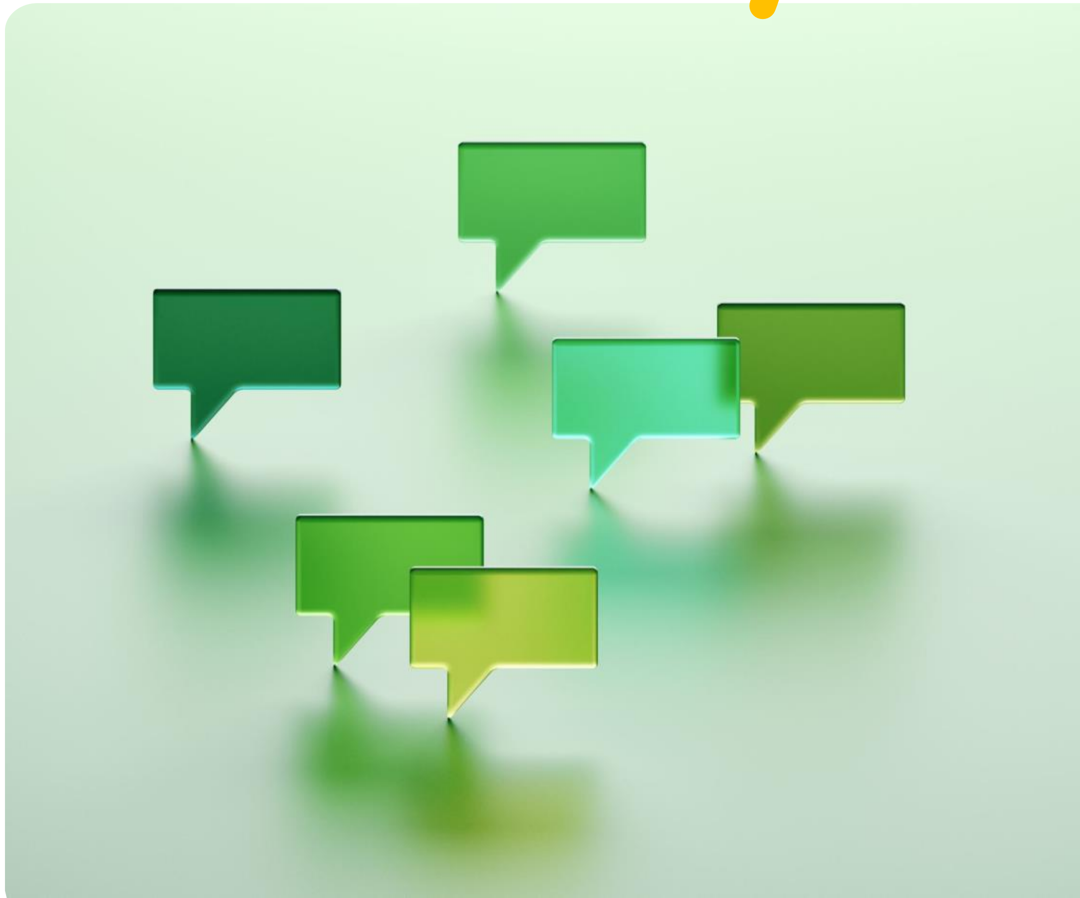
Data Frames

- To see how factors are represented in a data frame use the model.matrix() function.
- This function creates a set of dummy variables, one column for each level of a factor, represented with a 1 if that row contains that level and a 0 if not.
  - newFactor <- **factor**(**c**("Pennsylvania", "New York", "New Jersey",
    + "New York", "Tennessee", "Massachusetts",
    + "Pennsylvania", "New York"))
    > **model.matrix**(~ newFactor - 1)

	newFactorMassachusetts	newFactorNew Jersey	newFactorNew York
1	0	0	0
2	0	0	1
3	0	1	0
4	0	0	1
5	0	0	0
6	1	0	0
7	0	0	0
8	0	0	1

Lists

- A List is a container that can hold various data types.
- A List can contain all the same data types, a mix of data types, data frames, or other lists.
- Lists are useful in creating multi dimensional objects in to one object
- To create a list we use the "list()" function
    - Ex. list("a", 2, theDataFrame)
- You can use the "names()" function to see the names of each of the elements in a list.



Lists

- You can also assign names to the elements of a list while creating the list
  - names(list5) <- c(TheDataFrame = data.frame", TheVector = 1:10, TheList = list3)
- To create an empty list of a certain size use "vector()"
  - (emptyList <- vector(mode="List", length = 4))

Lists

- Access an element of a list using the [ ] and specify the element number or name. This allows for accessing only one element at a time.
  - Ex. list5[[1]]
  - Ex. list5[["data.frame"]]
- You can use the "length()" function to determine the length of a list

Lists

- You can add elements to a list by using an index that doesn't exist
  - # add a fourth element, unnamed
    > list5[[4]] <- 2
    > **length**(list5)
    [1] 4
    > # add a fifth element, name
    > list5[["NewElement"]] <- 3:6
    > **length**(list5)
    [1] 5
    > **names**(list5)

    [1] "data.frame" "vector" "list" "" "NewElement"
- Best practice is to make your list as long as it needs to be and then fill it in using the appropriate indices.

Matrices

- Similar to data.frames in that it is rectangular with rows and columns except that every single element, regardless of column, must be the same type, most commonly all `numerics`.
- Also act similarly to `vectors` with element-by-element addition, multiplication, subtraction, division and equality.
- **nrow**, **ncol** and **dim** functions work just like they do for `data.frames`
- Matrices are created using the "matrix()" function
  - # create a 5x2 matrix
> A <- **matrix**(1:10, nrow=5)

Matrices

- Matrix multiplication is a commonly used operation in mathematics, requiring the number of columns of the left-hand matrix to be the same as the number of rows of the **right-hand** matrix.
  - `Matrix` **multiplication keeps the row names from the left** `matrix` **and the column names from the right** `matrix`**.**
- `Matrices` can also have row and column names by using the rownames() and colnames() functions
- There are two special `vectors`, `"letters"` and `"LETTERS"`, that contain the lower case and upper-case letters, respectively.

Arrays

- Arrays are multidimensional vectors.
- Arrays must be all of the same type
- Elements of an array are accessed using the square brackets [ ]
  - The first element is the row index
  - The second is the column index
  - Remaining elements are the outer dimensions
- Difference between arrays and matrix is that matrices are restricted to 2 elements and an array can be however many you want.